



Universitas
Esa Unggul

Smart, Creative and Entrepreneurial

PTI 412-Internet Of Things

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SESI 1 dan SESI 2

IoT Paradigm



IoT Paradigm

CSI 421-Internet Of Things
Universitas Esa Unggul

- IoT Paradigm
- **Semua Bahan mengacu kepada buku : The Internet of Things: Enabling Technologies, Platforms, and Use Cases [Pethuru Raj, Anupama C. Raman]**

Apa itu IoT

Internet of Things adalah jaringan perangkat fisik yang saling terhubung melalui internet untuk mengumpulkan dan bertukar data.

Contoh sederhana: Smartwatch, Smart Home, Kamera CCTV online

Mengapa IoT terdengar strategis?

- **IoT Mengarah pada Komputasi yang Lebih Cerdas**
- **IoT Menghadirkan Lingkungan yang Lebih Cerdas**

Sejarah dan Perkembangan IoT

Istilah pertama kali dikenalkan tahun 1999 oleh Kevin Ashton

Berkembang dari konsep M2M (Machine to Machine)

Didorong oleh perkembangan sensor, internet, dan cloud computing

Konsep Dasar IoT

Sensor & Aktuator: mengumpulkan dan bereaksi terhadap data

Konektivitas: perangkat terhubung via Wi-Fi, Bluetooth, dsb

Data Processing: lokal atau cloud

Interface Pengguna: kontrol via aplikasi atau web

Komponen IoT

Perangkat Fisik (sensor, aktuator, mikrokontroler)

Koneksi Internet

Platform IoT / Cloud

Aplikasi pengguna

Protokol komunikasi (MQTT, HTTP, CoAP)

Protokol Komunikasi #1

MQTT (Message Queuing Telemetry Transport)

Kelebihan:

- Ringan dan hemat bandwidth → cocok untuk perangkat IoT
- Efisien untuk koneksi tidak stabil (seperti jaringan GSM/LoRa)

Cara kerja:

- Perangkat mengirim data (publish) ke broker → broker meneruskan ke perangkat lain yang subscribe

Contoh penggunaan: Sensor suhu mengirim data ke broker, lalu aplikasi menampilkan suhu secara real-time.

Protokol Komunikasi #2

HTTP (Hypertext Transfer Protocol)

Kelebihan:

- Umum digunakan di web → mudah diintegrasikan dengan API/web service
- Simple dan langsung

Kekurangan:

- Lebih berat dibanding MQTT dan tidak cocok untuk koneksi lemah

Cara kerja: Perangkat mengirim request ke server, lalu server memberi response

Contoh penggunaan: NodeMCU mengirim data sensor ke server lewat API HTTP.

Protokol Komunikasi #3

CoAP (Constrained Application Protocol)

Jenis: Protokol ringan berbasis UDP

Dirancang khusus untuk: Perangkat dengan sumber daya terbatas (low power, low memory)

Fitur:

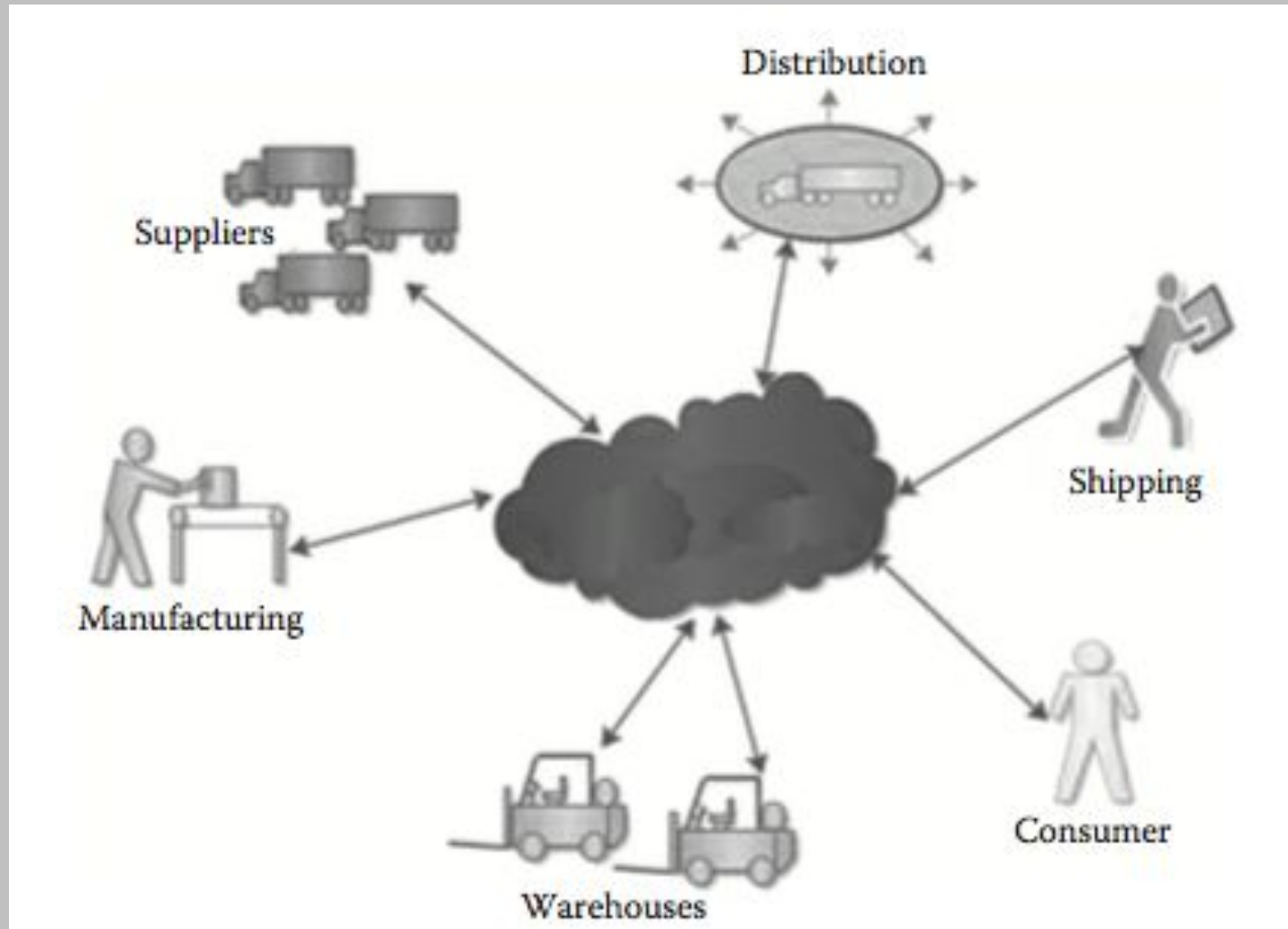
- Mirip HTTP tapi lebih ringan
- Bisa multicast (kirim ke banyak perangkat sekaligus)

Contoh penggunaan: Perangkat IoT di smart city (lampu jalan otomatis, sensor polusi)

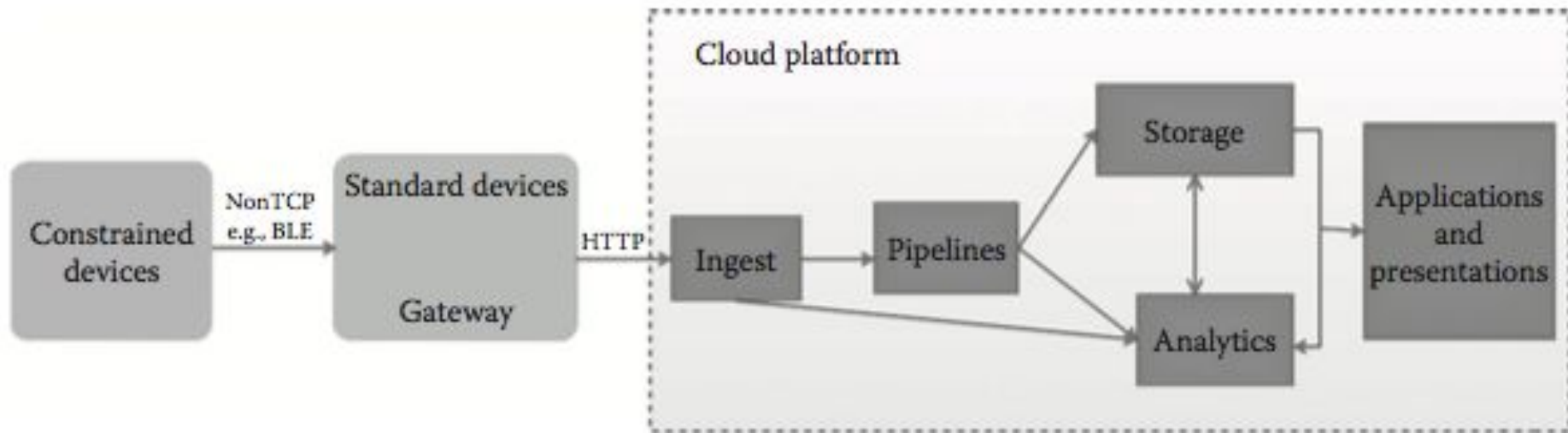
The Popular Applications

- ***Smart Energy***
- ***Smart Health Care***
- ***Smart Home Security***
- ***Smart Cargo Handling***
- ***Smart Traffic Management***
- ***Smart Inventory and Replenishment Management***
- ***Smart Cash Payment***
- ***Smart Tracking***
- ***Smart Displays***
- ***Smart Cargo Handling***
- ***Smart Traffic Management***
- ***Smart Inventory and Replenishment Management***
- ***Smart Cash Payment***
- ***Smart Tracking***
- ***Smart Displays***
- ***Smarter Manufacturing***
- ***Smart Asset Management***
- ***Smarter Retailing***

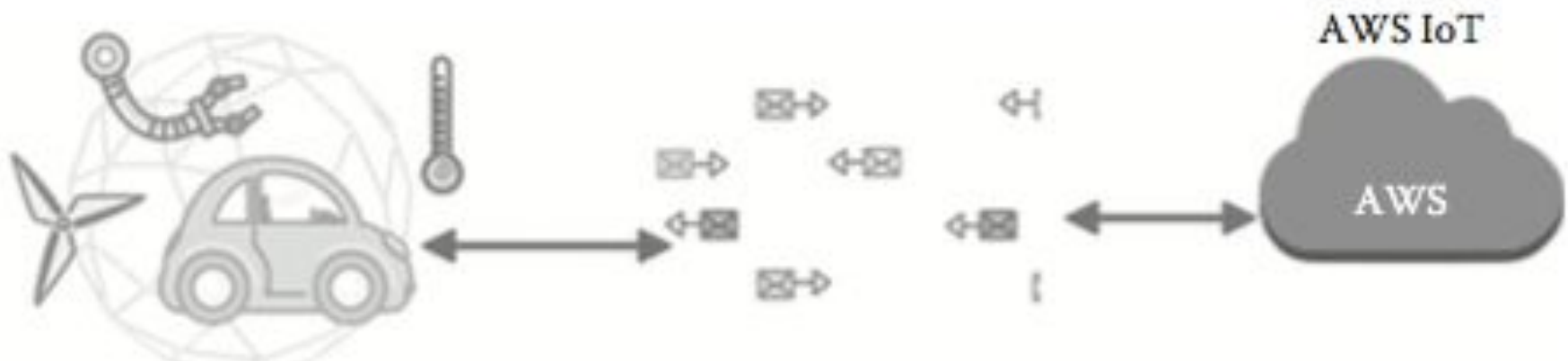
A cloud-based integrated environment



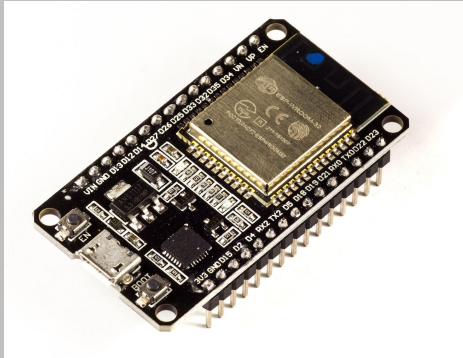
Google cloud platform architecture for stream processing from millions of intelligent devices.



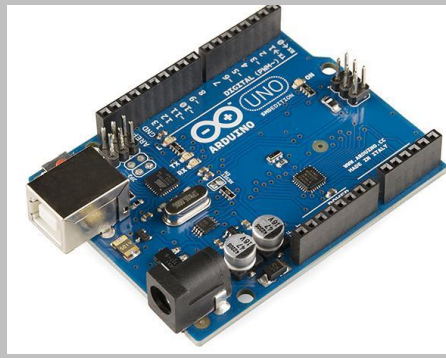
Device-to-cloud integration by AWS IoT.



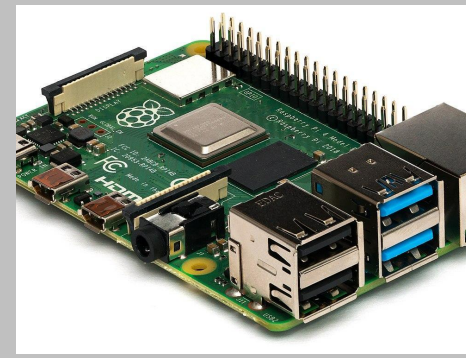
Jenis jenis Micro Controller yang umum digunakan



Esp 32



Arduino

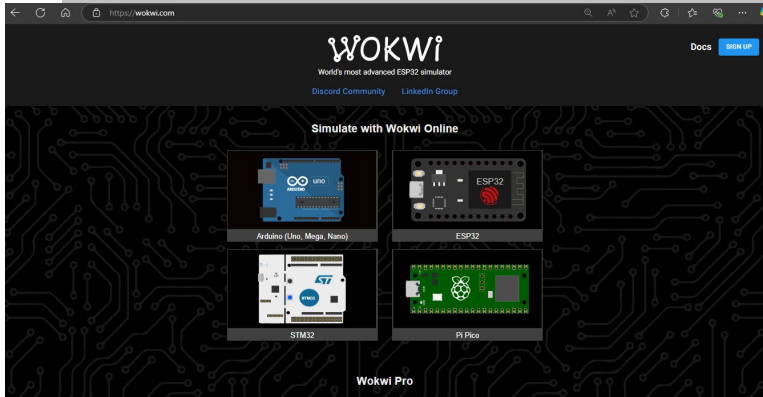


Raspberry Pi

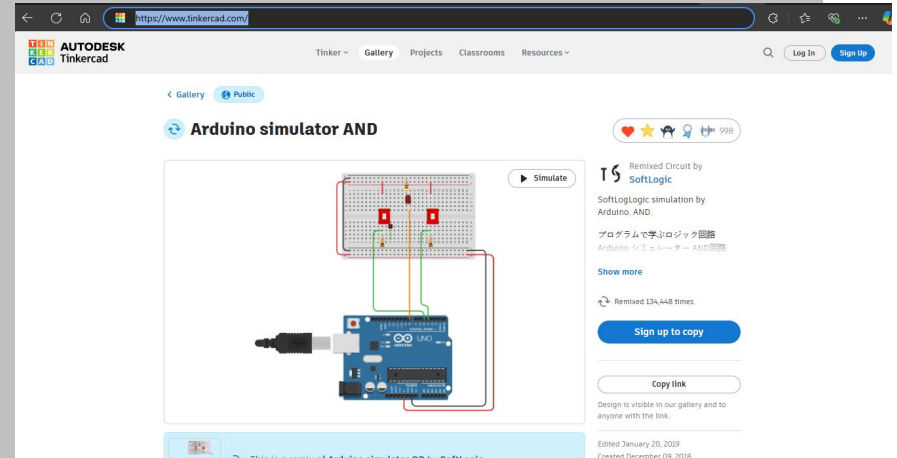
Tantangan IoT dan Ranah Penelitian

- *Arsitektur perangkat hemat energi*
- *Infrastruktur IoT yang elastis*
- *Protokol komunikasi yang sangat dioptimalkan*
- *Mekanisme deduplikasi dan kompresi data*
- *Keandalan data*
- *Keamanan perangkat*

Software yang digunakan untuk simulasi



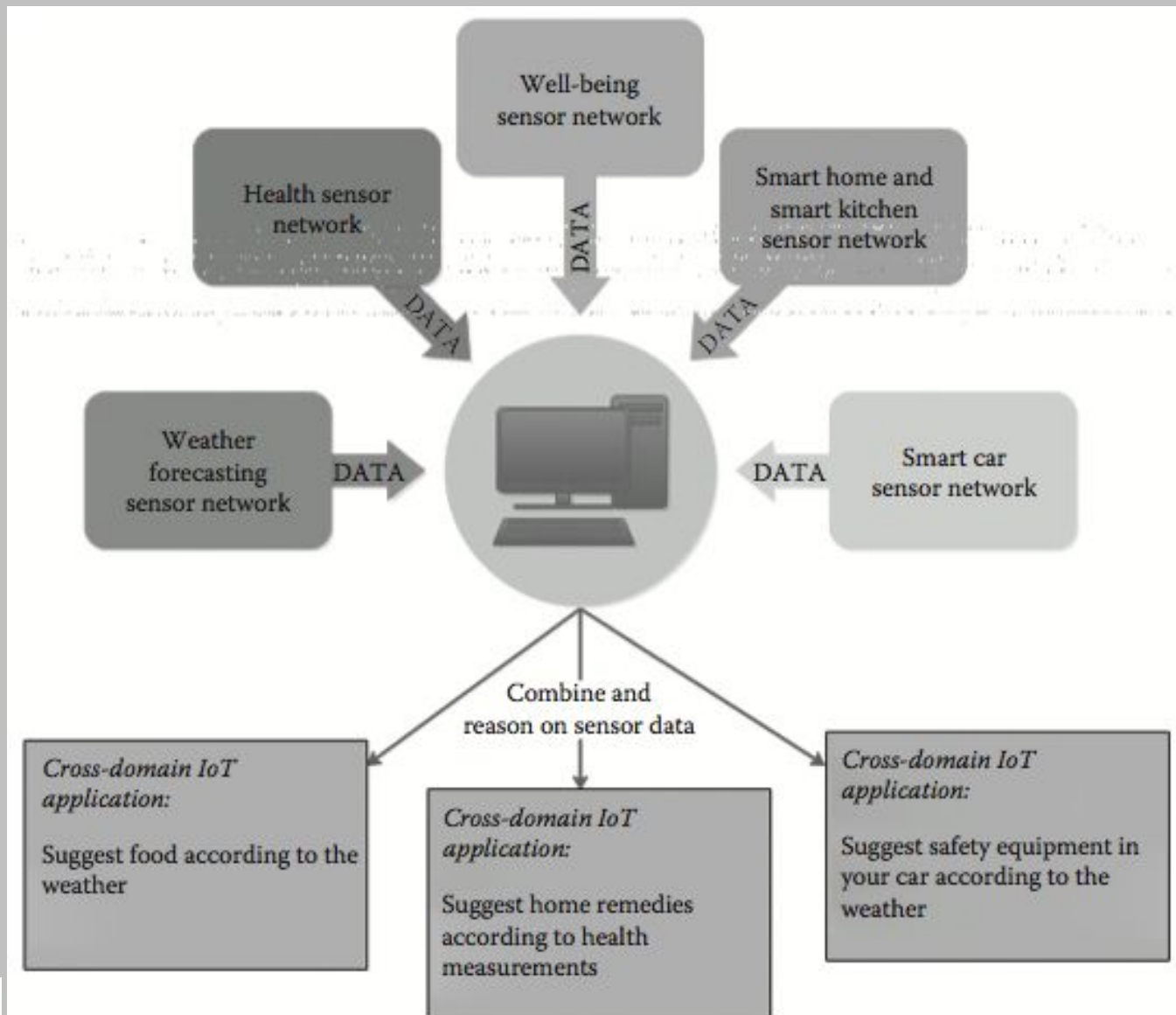
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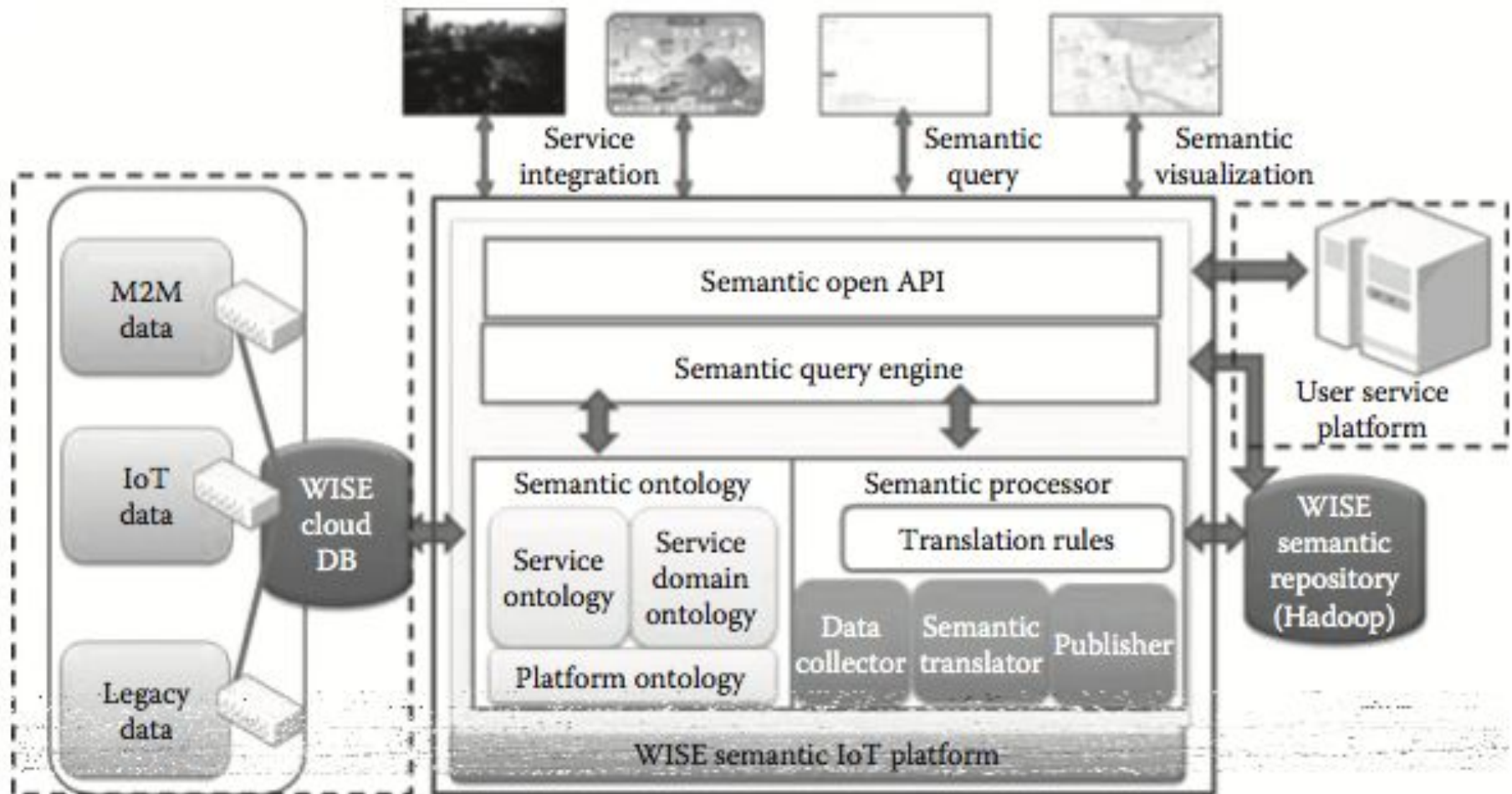
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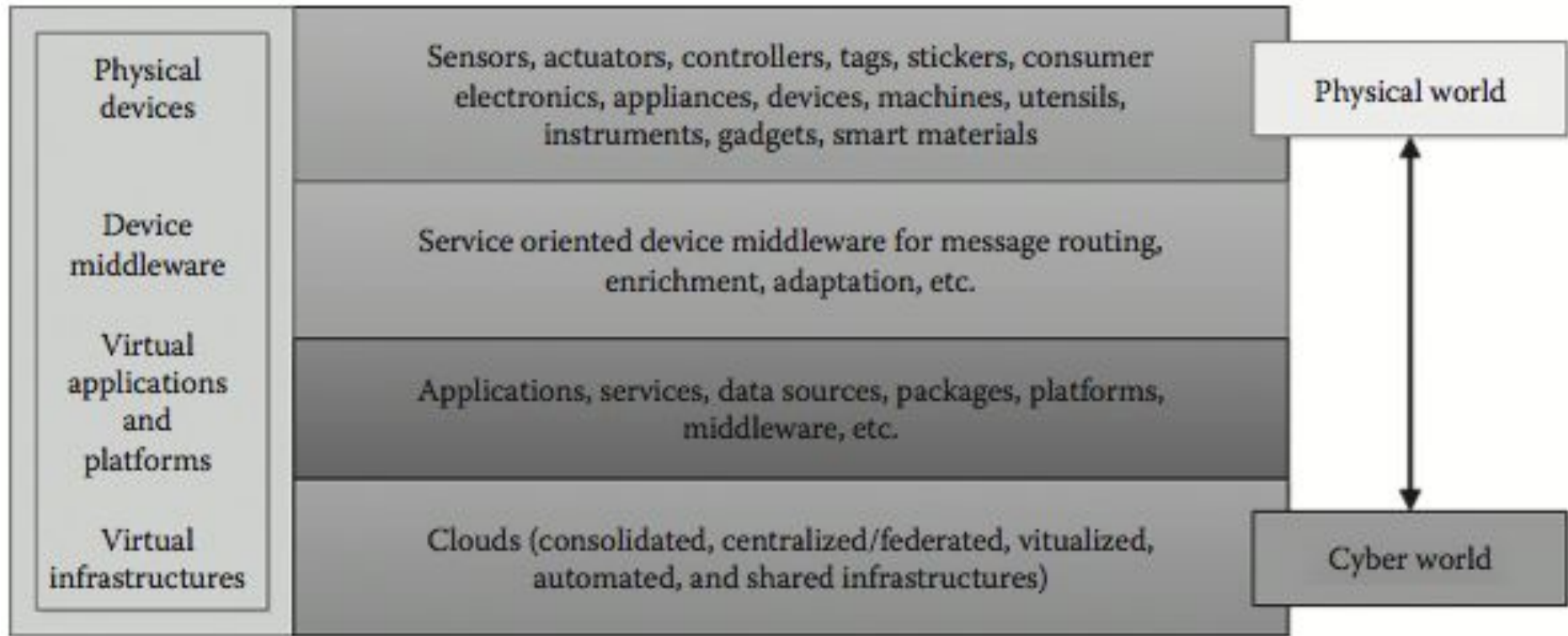
Device-to-device (D2D) integration



The semantic IoT platform architecture for weather information service engine



The integration of cyber and physical worlds.



The evolution of the Internet paradigm

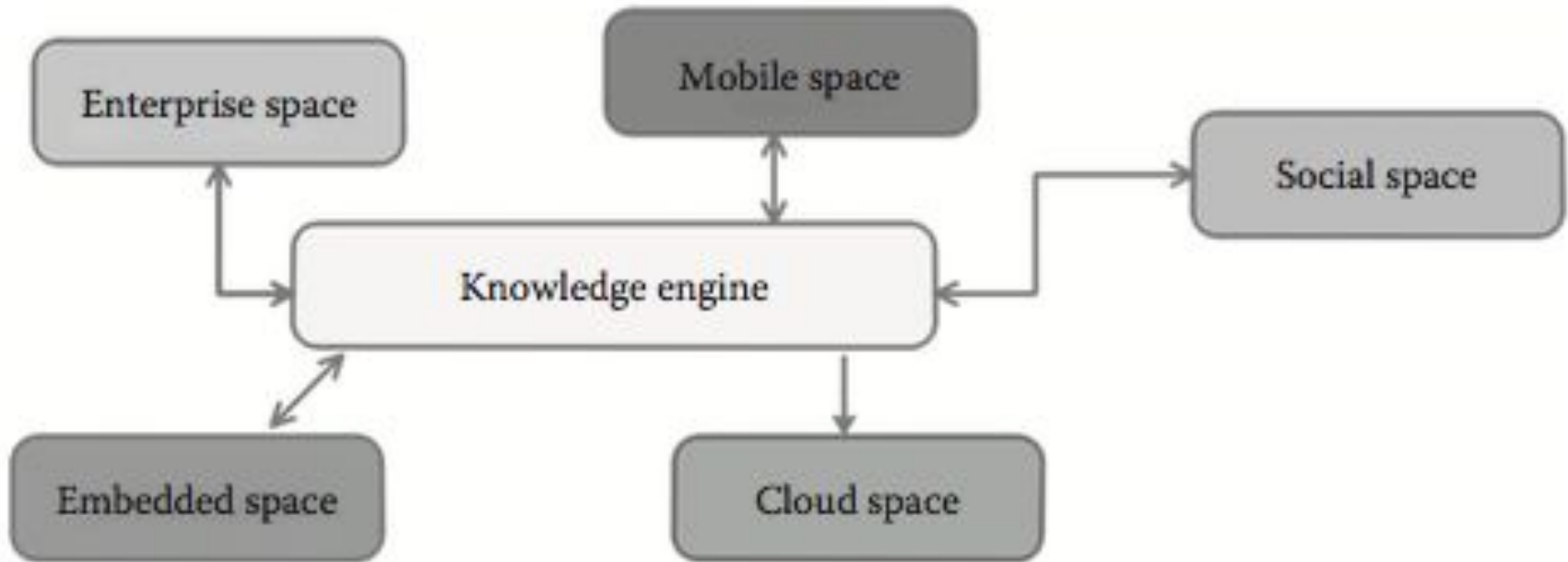
At the Connectivity and Infrastructure Level

- The Internet of Computers (IoC)
- The Internet of Devices (IoD)
- The Internet of Services (IoS)
- The Internet of Things (IoT)
- The Internet of Energy (IoE)

At the Content and Service Level

- Web 1.0 (The simple web - Reading only)
- Web 2.0 (The social web - Reading and writing)
- Web 3.0 (The semantic web - The automated searching and delivering right and relevant info for human interpretation)
- Web 4.0 (The smart web - Real time extraction and delivery of actionable insights to human consumption)

The futuristic integrated environments.



The IoT: The Key Application Domains

- *New business possibilities*
- *Tending toward the insights as a service (IaaS) era*
- *Fresh revenue opportunities*
- *Automation at its peak*

The Industrial Internet of Things (IIoT)

- *Improving operational efficiency*
- *Optimizing assets*
- *Envisioning next-gen services*
- *Exploring fresh avenues for higher revenues*

Few sectors wherein the IoT is bound to make waves of distinct automation:

- *Automotive industry*
- *Energy industry*
- *Health care*
- *Industrial*
- *Retail*
- *Smart buildings*

Consumer IoT market are as follows:

- Revenue generation solutions
- An ecosystem of developers to create great apps
- Help at the design stage to make apps an integral part of a manufacturer's IoT strategy
- A global highly scalable cloud platform to handle connections with millions of connected devices
- Security solutions for authentication and data protection
- Compelling on-device and companion app store experience
- Web stores for ease of viewing IoT app catalog on regular devices
- Software updating and management solutions
- An app supply chain compatible with, and tested for, each device
- Integrations with the device and the manufacturer's back-end systems
- Support for the lifecycle of the device software